

# Distill the Geometry, Not the Tokens

## Text-Measure Kernel Distillation for Universal Text Embeddings

Method Proposal for an ICLR 2027 Submission

Author Name | Institution | email@example.com

Provisional method name: TMKD. All theorem statements below are research targets, not completed claims.

**Core hypothesis.** Cross-tokenizer embedding distillation is usually formulated as token correspondence or transport. We instead model teacher and student hidden states as piecewise-constant vector fields over the same raw-text measure space. The student is trained to preserve the teacher-induced similarity kernel over text positions. A common refinement of tokenizer partitions computes this objective exactly, without learned alignment, optimal transport, projection heads, token salience heuristics, or layer selection.

### Abstract

Large embedding teachers and compact encoder students often use incompatible tokenizers. Existing cross-tokenizer knowledge distillation methods address this mismatch through one-to-one token mapping, learned projections, attention selection, or optimal transport. These mechanisms can be effective, but introduce brittle correspondences, multiple hyperparameters, and substantial computational overhead. This proposal develops *Text-Measure Kernel Distillation* (TMKD), a single-loss framework that defines contextual representations on the raw-text continuum rather than on tokenizer-specific indices. Teacher and student tokenizations are treated as two partitions of a shared measure space. Their final hidden states induce piecewise-constant vector fields and cosine kernels on this space. The distillation objective is the expected squared distortion between the two kernels, evaluated exactly by the common refinement of the two partitions. The resulting method is invariant to independent orthogonal transformations and hidden dimensions, has a bounded natural scale, and requires only one global loss coefficient. The theoretical program aims to establish exact discretization, refinement invariance, boundary stability, and an upper bound from token-kernel distortion to sentence-level embedding-kernel distortion. Experiments will follow the teacher, student, datasets, and optimization protocol of EMO, while emphasizing controlled tokenizer perturbations, fixed-coefficient performance, robustness across teachers, and training efficiency.

## 1 Motivation and Research Question

State-of-the-art text embedding models increasingly use heterogeneous architectures and tokenizers. Distilling a decoder-derived embedding teacher such as LLM2Vec into a compact BERT student therefore requires knowledge transfer across different sequence lengths, token boundaries, hidden dimensions, and internal coordinate systems. EMO addresses this problem by combining minimum-edit-distance token mapping, attention-structure alignment through centered kernel alignment, and importance-weighted optimal transport over final hidden states [1]. More recent cross-tokenizer work also emphasizes span aggregation or multi-space alignment [2, 3]. These methods expose a recurring tension:

- **Correspondence is underdetermined.** A token in one model may correspond to several subtokens, a partial morpheme, or no exact lexical unit in the other model.
- **Coordinates are not identifiable.** Embedding tasks depend primarily on similarities and distances, while direct hidden-state alignment penalizes arbitrary changes of basis.
- **Robust methods become over-parameterized.** Alignment choices introduce top- $m$ , layer maps, transport regularization, importance weighting, temperatures, projections, and dataset-specific loss coefficients.
- **Universal embedding quality is geometric.** Classification, sentence-pair inference, and semantic textual similarity all depend on how examples are organized in representation space, not on token identity alone.

The central research question is:

*Can cross-tokenizer embedding distillation be defined directly as preservation of representation geometry on the shared raw text, using a single objective that remains competitive without dataset-specific tuning?*

**Desired research outcome.** The project should produce a method whose scientific contribution is a change in the mathematical object being distilled, rather than an additional alignment module. Success requires competitive standard performance, substantially flatter hyperparameter sensitivity, and graceful degradation as tokenizer mismatch is increased under controlled perturbations.

## 2 Problem Setup

Let  $x$  be a raw text instance and let the teacher  $T$  and student  $S$  tokenize it into interval partitions

$$\mathcal{P}_x^T = \{I_{x,1}^T, \dots, I_{x,n_T}^T\}, \quad \mathcal{P}_x^S = \{I_{x,1}^S, \dots, I_{x,n_S}^S\}. \quad (1)$$

Each interval is represented by half-open character offsets  $[a, b)$ . Special tokens, padding, and positions without valid offsets are excluded. Let  $h_{x,i}^T \in \mathbb{R}^{d_T}$  and  $h_{x,j}^S \in \mathbb{R}^{d_S}$  denote final-layer contextual states. The hidden dimensions may differ.

The conventional approach seeks a cross-model map between token indices, representations, or probability distributions. TMKD instead treats tokenization as a numerical discretization of a shared base domain.

**Definition 1** (Raw-text measure space). *For each text  $x$ , define  $\Omega_x = [0, 1]$  with normalized length measure  $\mu_x$ . A character interval  $[a, b)$  receives mass  $(b - a)/|x|$ . Implementations may equivalently use Unicode codepoint or byte offsets, provided both tokenizers are mapped to the same underlying indexing convention.*

**Definition 2** (Contextual vector field). *The normalized teacher field is*

$$f_T(x, u) = \frac{h_{x,i}^T}{\|h_{x,i}^T\|_2} \quad \text{when } u \in I_{x,i}^T, \quad (2)$$

and  $f_S(x, u)$  is defined analogously. Each field is piecewise constant on its own tokenizer partition and maps into a unit sphere of a potentially different dimension.

## 3 Proposed Method: Text-Measure Kernel Distillation

### 3.1 Tokenizer-independent semantic kernel

For model  $M \in \{T, S\}$ , define the contextual similarity kernel

$$K_M((x, u), (x', v)) = \langle f_M(x, u), f_M(x', v) \rangle. \quad (3)$$

The kernel compares two raw-text positions and is independent of token indices, hidden dimension, and feature basis. It captures both local within-sentence relations and cross-sentence geometry.

The population distillation objective is

$$\mathcal{L}_{\text{TMKD}} = \mathbb{E}_{\substack{x, x' \sim \mathcal{D} \\ u \sim \mu_x, v \sim \mu_{x'}}} [K_T((x, u), (x', v)) - K_S((x, u), (x', v))]^2. \quad (4)$$

This is the expected squared distortion of teacher edge weights under the student geometry. Since cosine values lie in  $[-1, 1]$ , the normalized objective satisfies  $0 \leq \mathcal{L}_{\text{TMKD}} \leq 4$ .

### 3.2 Exact computation by common refinement

For each text, collect and sort all boundaries from both tokenizer partitions. Adjacent boundaries define the common-refinement atoms

$$\mathcal{R}_x = \{R_{x,1}, \dots, R_{x,m_x}\}, \quad (5)$$

where every atom lies inside exactly one teacher token and one student token. If  $t(x, r)$  and  $s(x, r)$  are those token indices, define

$$z_{x,r}^T = \frac{h_{x,t(x,r)}^T}{\|h_{x,t(x,r)}^T\|_2}, \quad z_{x,r}^S = \frac{h_{x,s(x,r)}^S}{\|h_{x,s(x,r)}^S\|_2}. \quad (6)$$

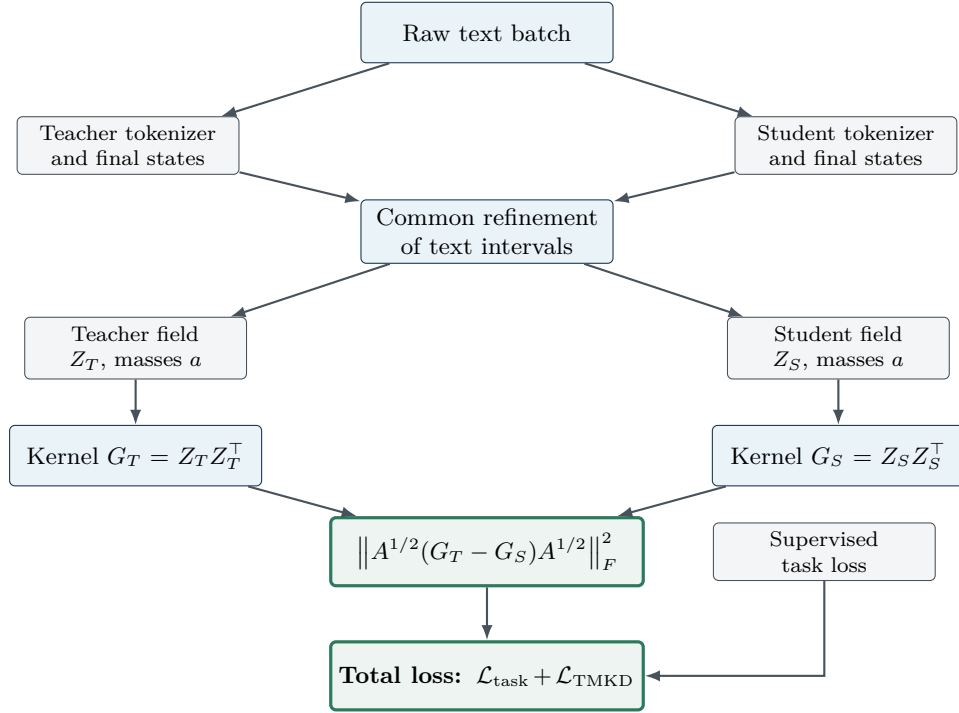


Figure 1: Overview of TMKD. Teacher and student tokenizations are lifted to a shared raw-text refinement. Only their induced kernels are compared; hidden coordinates are never aligned directly.

For a minibatch  $\mathcal{B} = \{x_1, \dots, x_B\}$ , each sentence receives equal total mass, while its atoms receive length-proportional mass:

$$a_{x,r} = \frac{1}{B} \frac{|R_{x,r}|}{\sum_{q=1}^{m_x} |R_{x,q}|}, \quad \sum_{x \in \mathcal{B}} \sum_r a_{x,r} = 1. \quad (7)$$

Concatenate the atom representations into  $Z_T \in \mathbb{R}^{N \times d_T}$  and  $Z_S \in \mathbb{R}^{N \times d_S}$ , where  $N = \sum_x m_x$ , and construct

$$G_T = Z_T Z_T^\top, \quad G_S = Z_S Z_S^\top, \quad A = \text{diag}(a_1, \dots, a_N). \quad (8)$$

The finite-batch objective is

$$\mathcal{L}_{\text{TMKD}}^{\mathcal{B}} = \left\| A^{1/2}(G_T - G_S)A^{1/2} \right\|_F^2 = \sum_{p,q} a_p a_q (G_{T,pq} - G_{S,pq})^2. \quad (9)$$

Because both vector fields are constant on each refinement atom, eq. (9) is exact quadrature of eq. (4) under the empirical batch distribution. Common refinement is therefore not a learned alignment module or heuristic correspondence rule; it is a deterministic integration procedure.

### 3.3 Training objective

The final student objective is intentionally minimal:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{TMKD}}^{\mathcal{B}}. \quad (10)$$

The main experiments will use a single global value  $\lambda = 1$  across all datasets, tasks, and teacher models. No dataset-specific search is used for the reported primary result. A small fixed grid is evaluated only as a robustness analysis, not for model selection.

### 3.4 Why the full batch kernel matters

A within-sentence Gram loss only constrains relations among positions inside the same text. Universal embeddings, however, are evaluated through relations between different examples. Concatenating all refinement atoms in a minibatch causes eq. (9) to include:

- relations within a single sentence;
- relations across the two members of a sentence pair;
- relations across unrelated examples in the minibatch.

No additional intra/inter weighting coefficient is introduced. The same probability measure generates all pair weights.

### 3.5 Graph interpretation

Each batch induces two weighted complete graphs on a shared node set. Nodes are refinement atoms with masses  $a_p$ , and edge weights are cosine similarities. The loss is the mean-squared edge distortion

$$\mathbb{E}_{p,q \sim a} [w_T(p, q) - w_S(p, q)]^2. \quad (11)$$

This provides a graph-geometric interpretation without choosing a neighborhood size, edge threshold, kernel bandwidth, graph Laplacian, or spectral rank.

## 4 Theoretical Program

The theory is intended to explain why the loss is the correct invariant object and how token-level kernel preservation controls sentence-level geometry. The statements below are targets for formal proof and refinement.

**Proposition 1** (Exact quadrature). *For piecewise-constant teacher and student fields induced by finite tokenizer partitions, the common-refinement sum in eq. (9) equals the continuous kernel-distortion integral in eq. (4) under the empirical batch measure.*

**Target Theorem 1** (Geometry identification modulo partial isometry). *Suppose the teacher and student kernels agree almost everywhere on the data support. Then there exists an inner-product-preserving linear map between the closures of the spans of the teacher and student fields that maps one field to the other almost everywhere. Consequently, zero kernel distortion identifies representation geometry up to unused dimensions and independent orthogonal coordinate systems.*

This result formalizes why no projection between  $d_T$  and  $d_S$  is required. The method preserves what cosine-based downstream decisions observe while ignoring non-identifiable coordinates.

**Target Theorem 2** (Tokenizer refinement invariance and boundary stability). *If a token interval is split into subintervals whose contextual states equal the original token state, the induced vector field and TMKDobjective are unchanged. More generally, if two fields differ only on a subset of raw-text measure at most  $\delta$ , then the change in bounded kernel distortion is at most  $C\delta$  for a universal constant  $C$ .*

The exact part captures invariance to purely representationally redundant token splitting. The stability part targets robustness to small token-boundary perturbations and imperfect offset conventions.

**Target Theorem 3** (Sentence-kernel domination). *Define the measure-pooled sentence representation*

$$e_M(x) = \int_{\Omega_x} f_M(x, u) d\mu_x(u), \quad (12)$$

*and the corresponding sentence kernel  $k_M(x, x') = \langle e_M(x), e_M(x') \rangle$ . Then*

$$\mathbb{E}_{x, x'} [ |k_T(x, x') - k_S(x, x')|^2 ] \leq \mathcal{L}_{\text{TMKD}}. \quad (13)$$

The proof follows by expressing the sentence kernel as the double integral of the position kernel and applying Jensen’s inequality. This provides the core bridge from token-level geometric preservation to universal sentence-embedding geometry.

**Target Corollary 1** (Similarity-order stability). *If the teacher’s sentence-similarity margin between two candidate pairs is larger than twice a uniform student kernel error bound, their relative order is preserved. A probabilistic version can be derived from eq. (13) for most pairs under the data distribution.*

**Optional extension.** If time permits, the project will derive a finite-sample concentration result for the minibatch estimator of eq. (4). This would quantify the effect of EMO’s small batch size and clarify when blockwise exact computation is preferable to pair sampling.

## 5 Implementation

### 5.1 Algorithm

```

Input: raw texts x[1:B], frozen teacher T, trainable student S
1. Run T and S with final hidden states and tokenizer offsets.
2. For each text, form the sorted union of teacher/student boundaries.
3. For every refinement interval:
   - copy the normalized teacher state of its containing teacher token;
   - copy the normalized student state of its containing student token;
   - assign length mass / (batch size * valid text length).
4. Concatenate all atom states into Z_T and Z_S.
5. Compute G_T = Z_T Z_T^T and G_S = Z_S Z_S^T.
6. Compute L_TMKD = sum_{p,q} a_p a_q (G_T[p,q] - G_S[p,q])^2.
7. Update the student with L_task + L_TMKD.

```

The teacher is always evaluated under `no_grad`. The common refinement and kernel loss are used only during training; inference is identical to the baseline student.

### 5.2 Computational complexity

For  $N$  total refinement atoms in a batch, explicit kernel computation costs

$$O(N^2(d_T + d_S)) \quad (14)$$

and  $O(N^2)$  memory. This is practical for the short-to-medium sequences and batch size used by EMO. Longer inputs can use exact blockwise accumulation without changing the objective. Block size is an implementation constraint, not a statistical or modeling hyperparameter.

### 5.3 Numerical details

- Add a fixed machine-stability constant to vector normalization only; this is not tuned.
- Compute kernel products in mixed precision but accumulate the weighted squared loss in FP32.
- Remove padding and special-token intervals before normalizing sentence mass.
- Preserve the baseline task head and pooling strategy for the main comparison. A theory-aligned length-pooled variant is reported only as a controlled ablation applied to all methods.

## 6 Experimental Plan

### 6.1 Primary setting

The main experiment follows EMO as closely as possible [1]:

- **Teacher:** LLM2Vec-Mistral 7B, with BGE-M3 as a teacher-robustness setting [4, 5].
- **Student:** BERT-base-uncased.

| Baseline                       | Purpose                                                         |
|--------------------------------|-----------------------------------------------------------------|
| <b>Student fine-tuning</b>     | Measures gain from any teacher signal.                          |
| <b>Direct hidden KD</b>        | Tests whether coordinate matching is sufficient.                |
| <b>MinED</b>                   | Tests one-to-one lexical alignment.                             |
| <b>DSKD</b>                    | Tests learned cross-space projection/alignment.                 |
| <b>MultiLevelOT</b>            | Tests transport-based sequence alignment.                       |
| <b>EMO</b>                     | Primary embedding-specific CTKD baseline.                       |
| <b>SRA-style span geometry</b> | Controls for raw-offset span aggregation and pairwise geometry. |
| <b>Within-text TMKD</b>        | Ablates cross-sentence kernel relations.                        |
| <b>Full TMKD</b>               | Proposed method with one global coefficient.                    |

Table 1: Planned comparisons. Methods unavailable for a task because they require logits will be marked explicitly rather than silently omitted.

- **Text classification:** Patent, IMDb, Banking77.
- **Sentence-pair classification:** SciTail, ConTRoL-NLI, ANLI-R2.
- **Semantic textual similarity:** STS-B, STS12, SICK.
- **Optimization:** the same epoch budget, learning rate, batch size, task losses, heads, and evaluation metrics as the reproduced EMO setting.

The proposal does not expand immediately to retrieval. The claim is universal representation transfer over the same non-retrieval task families evaluated by EMO.

## 6.2 Baselines

### 6.3 Main evaluation axes

#### 6.3.1 Standard downstream performance

Report task-standard metrics and the mean across datasets. In addition, report teacher-gap closure

$$\text{Retention} = \frac{\text{Score}_{S+KD} - \text{Score}_S}{\text{Score}_T - \text{Score}_S}, \quad (15)$$

when higher is better and the denominator is positive. Mean retention, worst-task retention, and variation across seeds are more informative than an aggregate score alone.

#### 6.3.2 Controlled tokenizer mismatch

Construct tokenizer perturbation families at increasing severity:

- deterministic token splitting;
- adjacent-token merging;
- vocabulary reduction or tokenizer replacement;
- punctuation and whitespace boundary perturbations.

The main robustness statistic is the performance-degradation slope as mismatch increases. This experiment distinguishes a genuinely tokenizer-stable principle from a method that simply performs well at one teacher-student pair.

#### 6.3.3 Fixed-coefficient robustness

The primary table always uses  $\lambda = 1$ . A sensitivity plot over

$$\lambda \in \{0.25, 0.5, 1, 2, 4\} \quad (16)$$

reports mean, worst-case score, and dispersion. The grid is not used to select a per-dataset result. A broad performance plateau is a central success criterion.

### 6.3.4 Teacher and tokenizer controls

Evaluate at least one teacher with a substantially different tokenizer and one same-tokenizer or near-same-tokenizer control. The method should remain beneficial when mismatch is absent; otherwise, the gain may be only an offset-alignment effect rather than general geometric transfer.

### 6.3.5 Efficiency

Measure wall-clock training time per batch, peak memory, and inference cost. EMO’s OT component creates meaningful training overhead, while TMKD replaces transport iterations and attention-layer selection with two matrix products. Inference cost must be identical to the base student.

## 6.4 Minimal ablations

To preserve the simplicity claim, the planned ablation set is deliberately small:

1. **Index matching vs. common refinement:** Is the shared raw-text measure necessary?
2. **Pointwise vs. kernel geometry:** Is coordinate-free relational matching necessary?
3. **Within-text vs. full-batch kernel:** Do cross-sentence relations explain universal embedding transfer?
4. **Uniform vs. length mass:** Does exact measure weighting matter?
5. **Same tokenizer vs. cross tokenizer:** Is the method broadly geometric rather than mismatch-specific?

No kernel family, graph neighborhood, layer-selection strategy, learned salience, or auxiliary loss is introduced unless the core hypothesis fails and the project is explicitly reframed.

## 7 Expected Contributions

A successful paper will make three claims:

1. **Conceptual:** Cross-tokenizer representation distillation can be formulated on a shared raw-text measure space, making token alignment a discretization issue rather than a correspondence-learning problem.
2. **Theoretical:** Kernel distortion is coordinate-free, exactly computable by common refinement, stable to tokenizer refinement, and controls sentence-level embedding-kernel distortion.
3. **Empirical:** A single loss with one fixed global coefficient is competitive with or better than multi-component CTKD methods, while being more stable under tokenizer mismatch, teacher replacement, and coefficient variation.

The intended contribution is not merely a faster alignment trick. It is a principled definition of which representation object should be transferred when teacher and student tokenizations disagree.

## 8 Risks, Falsification Tests, and Kill Criteria

The project should be stopped or substantially reframed if: (i) the full kernel does not improve on a within-sentence Gram loss; (ii) gains vanish under fixed  $\lambda$ ; (iii) robustness under artificial split/merge perturbations is not better than alignment baselines; (iv) improvements occur for only one teacher-student pair; or (v) competitive performance requires multiple auxiliary losses, learned weighting, or dataset-specific search.

| Risk                                                           | Diagnostic                                                          | Response                                                                                                            |
|----------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Common-refinement atoms are not semantically meaningful</b> | Kernel loss falls without sentence-kernel or downstream improvement | Reassess the vector-field abstraction; do not add salience heuristics by default.                                   |
| <b>Quadratic kernel is too expensive</b>                       | Memory or time exceeds EMO substantially                            | Use exact blockwise accumulation; preserve the same objective.                                                      |
| <b>Cross-sentence relations are noisy at batch size four</b>   | Full-batch loss underperforms within-text loss                      | Analyze estimator variance; consider larger gradient accumulation only if protocol remains fair.                    |
| <b>Fixed <math>\lambda</math> is unstable</b>                  | Different tasks require sharply different optima                    | The simplicity thesis is falsified; do not report per-dataset tuning as the main method.                            |
| <b>Gain comes only from off-sets</b>                           | Same-tokenizer control shows no benefit                             | Reframe as tokenizer alignment rather than universal geometry, likely targeting a different venue.                  |
| <b>Overlap with span geometry is too strong</b>                | SRA-style baseline matches the method                               | Novelty requires the continuous kernel formulation, cross-sentence theory, and robustness evidence; otherwise stop. |

Table 2: Primary risks and planned falsification tests.

## 9 Milestones

| Phase               | Deliverable                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>    | Exact offset refinement, blockwise kernel loss, and gradient/unit tests on synthetic token splits.               |
| <b>Reproduction</b> | Reproduce student and EMO baselines on one classification, one pair-classification, and one STS dataset.         |
| <b>Core test</b>    | Compare within-text and full-batch TMKD with fixed $\lambda = 1$ . Apply the first kill criteria.                |
| <b>Robustness</b>   | Controlled tokenizer perturbations, coefficient plateaus, teacher replacement, and same-tokenizer controls.      |
| <b>Theory</b>       | Formalize exact quadrature, partial-isometry identification, boundary stability, and sentence-kernel domination. |
| <b>Full paper</b>   | Complete nine-dataset study, efficiency analysis, minimal ablations, and reproducible release.                   |

## 10 Provisional Paper Thesis

**Thesis.** Teacher and student tokenizers are different partitions of the same underlying text. The invariant object shared across those partitions is not a token correspondence, but the contextual similarity kernel induced over raw-text positions. Distilling this kernel yields a coordinate-free and tokenizer-stable objective whose finite implementation is exact, whose scale is naturally bounded, and whose preservation provably controls sentence-level embedding geometry.

**Provisional title:** *Distill the Geometry, Not the Tokens: Text-Measure Kernel Distillation for Universal Text Embeddings.*

## References

- [1] Minh-Phuc Truong, Hai An Vu, Tu Vu, Nguyen Thi Ngoc Diep, Linh Ngo Van, Thien Huu Nguyen, and Trung Le. EMO: Embedding Model Distillation via Intra-Model Relation and Optimal Transport Alignments. In *Proceedings of EMNLP*, 2025.

- [2] Quoc Phong Dao, Hoang Son Nguyen, Pham Khanh Chi, Tung Nguyen, Linh Ngo Van, Nguyen Thi Ngoc Diep, and Trung Le. SRA: Span Representation Alignment for Large Language Model Distillation. In *Proceedings of ACL*, 2026.
- [3] Duc Trung Vu, Pham Khanh Chi, Dat Phi Van, Linh Ngo Van, Dinh Viet Sang, and Trung Le. DWA-KD: Dual-Space Weighting and Time-Warped Alignment for Cross-Tokenizer Knowledge Distillation. In *Findings of EACL*, 2026.
- [4] Parishad BehnamGhader, Vaibhav Adlakha, Marius Mosbach, Dzmitry Bahdanau, Nicolas Chapados, and Siva Reddy. LLM2Vec: Large Language Models Are Secretly Powerful Text Encoders. In *Proceedings of ACL*, 2024.
- [5] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. 2024. arXiv:2402.03216.
- [6] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of Neural Network Representations Revisited. In *Proceedings of ICML*, 2019.
- [7] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the Knowledge in a Neural Network. 2015. arXiv:1503.02531.
- [8] Niklas Muennighoff, Nouamane Tazi, Loic Magne, and Nils Reimers. MTEB: Massive Text Embedding Benchmark. In *Proceedings of EACL*, 2023.